Mini Project On

GeoHappen: A Geo-Location Based Event Finder

Submitted in partial fulfillment of the requirements for the award of the degree of

Minor

In

Computer Science & Engineering

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October 2025

CERTIFICATE

This is to certify that the mini project report entitled ***"GeoHappen: A Geo-Location Based Event Finder"*** *is a bonafide record of the work done by* ***Aleena Shajan (U2201023), Anagha M Anil (U2201031), Angelina Mary Giby (U2201034), Architha B (U2201051)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of BTech Minor in "Computer Science & Engineering" during the academic year 2025-2026.*

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ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Rev. Dr. Jaison Paul Mulerikkal CMI**, Principal of RSET, and "Dr.Preetha K G", Head of the Department of "Computer Science Engineering" for providing us with the opportunity to undertake our project, "GeoHappen: A Geo-Location Based Event Finder".

We are highly indebted to our project coordinator, **Ms. Dincy Paul** , Assistant Professor, Department of Computer Science & Engineering, for her valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express our sincere gratitude to our project guide **Mr. Ajith S** for his patience and all the priceless advice and wisdom he has shared with us.

Last but not the least, we would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

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Abstract

This project, titled *GeoHappen: A Geo-Location Based Event Finder Web Application*, focuses on developing an interactive platform that allows users to discover, create, and engage with events based on their geographical location. The system integrates modern web technologies such as JavaScript , Node.js, Express.js, Firebase Authentication, Firestore, and Google Maps API to provide a seamless and responsive experience. Users can register, log in, add events, filter by category or date, and receive personalized recommendations based on their interests. Unlike existing event-finder applications that rely on static listings or manual searches, *GeoHappen* introduces real-time map-based event visualization and intelligent recommendations using user preferences. This enhances accessibility, reduces search effort, and promotes greater participation in events. The project demonstrates how cloud integration and real-time data handling can be effectively combined to create a dynamic, location-aware platform that benefits both event organizers and participants.

Contents

List of Abbreviations

API - Application Programming Interface

GPS - Global Positioning System

HTML - HyperText Markup Language

CSS - Cascading Style Sheets

JS - JavaScript

DB - Database

UI - User Interface

UX - User Experience

HTTP - HyperText Transfer Protocol

JSON - JavaScript Object Notation

CRUD - Create, Read, Update, Delete

IDE - Integrated Development Environment

URL - Uniform Resource Locator

DOM - Document Object Model

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Chapter 1

Introduction

This chapter provides an overview of the project *GeoHappen: A Geo-Location Based Event Finder Web Application*. It introduces the motivation behind the system, its objectives, scope, and the technological foundation on which it is built. The chapter also outlines the key challenges faced during development, assumptions made during implementation, and the societal as well as industrial relevance of the project. Overall, this chapter sets the context for understanding how the proposed system addresses the problem of discovering and managing events efficiently by providing a map-based visualization of all ongoing and upcoming events using modern web and cloud technologies.

1.1 Background

In the digital era, events such as college fests, workshops, seminars, cultural gatherings, and concerts are increasingly promoted online. However, users often struggle to find and keep track of events taking place around them due to the lack of a centralized platform. GeoHappen is a web application that provides an interactive map-based interface where users can view all events created by organizers. The map also displays the user's current location, allowing them to visually identify which events are closest to them. By integrating geolocation, Google Maps API, and Firebase Firestore, this system enables users to easily explore, filter, and interact with events. The platform bridges the gap between event organizers and participants, promoting greater community engagement and efficient information sharing.

1.2 Problem Definition

In today's digital era, events such as college fests, workshops, concerts, seminars, and meetups are often publicized on various online platforms. However, users still face challenges in finding events that are relevant to their interests and location. There is no single integrated platform that combines real-time map-based event visualization with user location awareness and intelligent recommendations. As a result, many potential participants miss events that match their preferences and schedules.

The core problems identified are as follows:

1. Users lack a unified platform to visualize and explore all events in real time through an interactive map.
2. Existing systems rarely use geolocation to display the user's position and surrounding events dynamically on a map.
3. Event organizers struggle to reach their target audience efficiently due to platform fragmentation.
4. Users must manually search through multiple sources (social media, posters, websites), which is time-consuming and inefficient.
5. There is limited provision for user-specific event recommendations based on interests or past activity.

1.3 Scope and Motivation

Scope: The project encompasses the design and development of a full-stack web application integrating geolocation and event data. It allows users to register, view, like events ,and interact with all available events displayed on an interactive map. The map highlights the user's current location, helping them visually identify events. Event organizers can add or delete events, while users receive recommendations based on their preferences and past interactions. The system is scalable and can be extended for large-scale event management.

Motivation: Many individuals miss out on events in their vicinity due to a lack of centralized, intelligent, and location-based event discovery systems. *GeoHappen* was motivated by the need to create a user-friendly, real-time event platform that enhances user experience and social interaction. By combining modern web technologies with cloud-based data handling, the project encourages digital community participation and efficient event dissemination.

1.4 Objectives

1. To develop a real-time, map-based event discovery web application with user location awareness.

2. To implement a user authentication and profile management system using Firebase Authentication.

3. To store and manage event data using Firebase Firestore in real time.

4. To integrate Google Maps and Geolocation API for interactive map-based event visualization.

5. To enable event filtering by date, category, and user preferences.

6. To generate event recommendations based on user likes and interests.

1.5 Challenges

Developing a real-time event discovery platform involves challenges such as handling accurate geolocation data, ensuring fast data retrieval, and dynamically updating map markers. Additionally, maintaining user privacy and seamless synchronization with Firebase requires efficient backend and database design.

1.6 Assumptions

1. Users have internet access and a device that supports geolocation services.

2. Event organizers provide accurate information regarding event details and location.

3. Users grant permission for location access so their current position can be displayed on the map, allowing them to visually identify events happening around them.

1.7 Societal / Industrial Relevance

GeoHappen has strong relevance in both societal and industrial contexts. For society, it fosters engagement by helping users explore events in their surroundings and promoting participation through an intuitive, map-based interface, enhancing participation in community activities, and promoting social interaction. In industry, the system can serve as a foundation for commercial event management platforms, local business promotions, and tourism applications that rely on location-based recommendations.

1.8 Organization of the Report

The report is organized into five chapters as follows:

- **Chapter 1 – Introduction:** Provides an overview of the project, background, problem definition, motivation, and objectives.
- **Chapter 2 – Literature Review:** Discusses existing event discovery systems, related web technologies, and research gaps identified.
- **Chapter 3 – System Design and Methodology:** Describes the overall system architecture, database design, MVC model, and development methodology adopted.
- **Chapter 4 – System Implementation:** Presents the implementation details of the GeoHappen web application, including frontend, backend, and API integration.
- **Chapter 5 – Results and Discussion:** Presents the obtained results, screenshots of the working system, and a detailed analysis of performance, usability, and system behavior.
- **Chapter 6 – Conclusion and Future Scope:** Summarizes the project outcomes, conclusions drawn, and possible future enhancements.

This chapter introduced the concept and motivation behind *GeoHappen*, along with its objectives, challenges, and relevance. The system aims to provide a smart, efficient, and user-centric approach to event discovery using modern web technologies and cloud-based solutions.

Chapter 2

Literature Survey

2.1 Introduction

A literature survey provides a comprehensive understanding of prior work related to geo-location-based systems, event recommendation engines, and context-aware applications. It forms the foundation for identifying technological advancements, research gaps, and potential directions for improvement. The reviewed works mainly focus on location-aware event discovery, intelligent management systems, and recommender frameworks that leverage geolocation and user interest data.

2.2 Review of Related Works

2.2.1 Navigating Event Landscapes: A Study of GPS-Based Event Discovery Systems

Saee Majlekar et al. (2024) proposed a GPS-based event discovery system that allows users to identify events dynamically through location tracking and filtering [1]. The system integrates geospatial data for efficient real-time recommendations and provides personalized suggestions based on user location. Although effective in identifying events, the study highlights limitations such as delayed real-time updates, dependency on continuous connectivity, and a lack of integration with advanced cloud-based synchronization tools. These limitations inspired the adoption of cloud databases such as Firebase in the proposed system **GeoHappen**.

2.2.2 Intelligent Event Finder and Management System

Afsar et al. (2021) presented an intelligent event management platform that connects users and organizers via cloud and IoT technologies [2]. The system efficiently manages event data and enables real-time communication between users and hosts. However, it lacked interactive visual elements such as dynamic mapping and real-time geolocation display. In **GeoHappen**, these shortcomings are addressed through Google Maps API integration and the use of Node.js and Firebase for responsive event updates.

2.2.3 Interest-Aware Location-Based Recommender System

AlBanna et al. (2016) introduced an interest-aware location-based recommender system that utilizes geo-tagged social media data to provide personalized recommendations [3]. Their model analyzed user behavior and preferences from social media platforms to suggest relevant points of interest (POIs). This study underscores the role of user preferences in improving the relevance of location-based suggestions. The proposed **GeoHappen** system extends this concept by incorporating both location proximity and event category preferences in its recommendation mechanism.

2.2.4 Context-Aware Group Recommendation for Point-of-Interests

Zhu et al. (2018) proposed a context-aware group recommender framework for POIs that considers user context, group influence, and location rationality [4]. The system applies group modeling to recommend suitable venues for multiple users simultaneously. While it effectively supports collaborative recommendations, its scope remains limited to static POIs rather than dynamic, time-bound events. The approach provides valuable insights for improving group-based event recommendations in systems like **GeoHappen**, especially for public or campus events.

2.2.5 Comparative Insights

The reviewed systems collectively demonstrate the evolution of location-based services—from basic GPS tracking to intelligent, context-aware recommendation engines. Table 2.1 summarizes their main contributions, limitations, and how **GeoHappen** addresses these gaps.

Table 1.1: Comparison of Existing Systems with the Proposed System (GeoHappen)

System / Study	Technology Used	Features Offered	Limitations	GeoHappen Advantages
Majlekar et al. (2024)	GPS, Cloud Database	Location-based event discovery and recommendations	Limited real-time synchronization and offline support	Real-time Firebase synchronization and responsive updates
Afsar et al. (2021)	IoT, Cloud, Web Integration	Event management and user-organization platform	Lacks interactive map and visualization tools	Interactive UI with Google Maps and instant event updates
AlBanna et al. (2016)	Geo-tagged Social Media, Recommender Systems	Interest-aware POI recommendation	Focused on social media data only, not dynamic events	Event-specific personalized recommendations based on user preferences
Zhu et al. (2018)	Context-aware Algorithms, Group Modeling	Group-based POI recommendations considering user context	Limited to static POIs and lacks temporal dynamics	Real-time event-based recommendations with time and location context
Proposed GeoHappen System	Node.js, Firebase, Google Maps API	Real-time event discovery, filtering, and recommendations across devices	–	Combines accuracy, cloud efficiency, and user personalization

2.3 Research Gaps Identified

From the literature reviewed, the following research and implementation gaps were identified:

- **Lack of real-time updates:** Many existing systems provide delayed event updates or require manual refresh.
- **Limited integration of mapping APIs:** Systems often lack real-time map visualization and distance-based filtering.
- **Insufficient personalization:** User preferences are not fully utilized in event-based systems, limiting recommendation quality.
- **Low scalability and cross-platform support:** Some applications are restricted to mobile platforms or small datasets.
- **No unified interaction between users and organizers:** Few existing systems provide a two-way communication framework.

2.4 Summary of Findings

From the above studies, it is evident that there is a strong demand for a unified, web-based system that enables location-aware event discovery with high responsiveness and user personalization. The proposed **GeoHappen** system bridges this gap by leveraging modern technologies such as Node.js, Firebase, and Google Maps API to offer an efficient, scalable, and interactive event discovery experience.

Chapter 3

System Design and Methodology

3.1 Overview

The system design of **GeoHappen** follows a modular and layered architecture to ensure scalability, maintainability, and efficient communication between client and server components. The design integrates geolocation-based services, real-time databases, and event management functionalities into a cohesive framework.

3.2 System Architecture

GeoHappen is built on a **Client–Server Model** that adopts the **Model–View–Controller (MVC)** architecture. This approach separates concerns, allowing independent development of the user interface, business logic, and data management.

- **View Layer (Client):** The frontend, built using HTML, CSS, and JavaScript, enables users to view and filter events, interact with the map, and access event recommendations. It uses Google Maps to display event markers based on geolocation.
- **Controller Layer (Application Logic):** Implemented using Node.js, this layer handles incoming requests, processes data, performs validation, and communicates with the model for data retrieval or updates.
- **Model Layer (Data):** This layer manages all data interactions with Firebase Firestore, storing and retrieving user and event data. It also integrates with the Google Maps and Geolocation APIs for location processing.

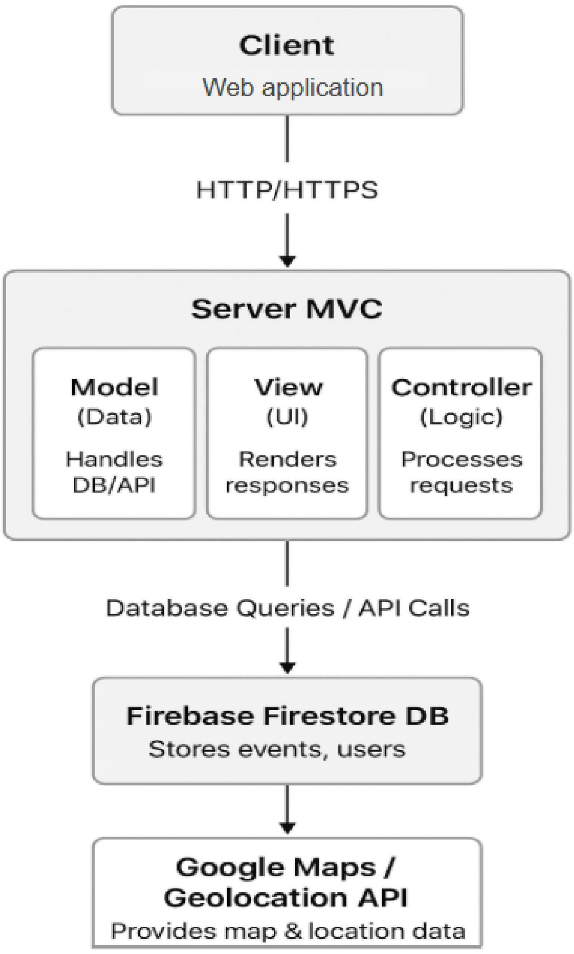

Figure 3.1: System Architecture of GeoHappen

3.2.1 Functional Blocks

1. **User Interface Module:** Displays events, event details, and map markers. Includes filtering options, login functionality, and a recommendation section.

2. **User Experience (UX) Module:** Handles responsiveness, smooth navigation, and fallback messages (e.g., when no events are found or GPS access fails).

3. **Database Module:** Stores user profiles, event data, and preferences in Firebase Firestore, providing real-time data synchronization.

4. **API Integration Module:** Uses:

   - Google Maps JavaScript API for event visualization.
   - Geolocation API for determining user position.

- Firebase SDK for authentication and data management.

3.3 Database Design

The database design of **GeoHappen**, a geolocation-based event finder web application, aims to store and manage data efficiently for users, events, and their spatial relationships. The system uses **Firebase Firestore** as the real-time database, but its conceptual design can also be represented using a relational model and ER diagram for better understanding.

The database is structured to support:

- Real-time event updates and retrieval
- Geolocation-based search
- User-event interaction and recommendations

3.3.1 Database Schema Design

The **database design diagram** represents the logical structure of how data is stored and related. It contains four main entities: *Users*, *Events*, *Categories*, and *Event_Tags*.

- **Users Table:**

  Stores information about registered users, including their name, email, password (hashed), and account creation timestamp. Each user can create or manage multiple events.

- **Events Table:**

  This is the central entity in the database. It stores key details such as event title, description, location, latitude, longitude, and event timings. Each event is linked to a specific *category* and *user (creator)*.

- **Categories Table:**

  Defines different types of events (e.g., Music, Workshop, Fest). Each category can have multiple events associated with it.

- **Event_Tags Table:**

  Helps classify or tag events based on additional attributes (e.g., "Free Entry," "Outdoor," "Online"). Each tag is linked to an event via a foreign key relationship.

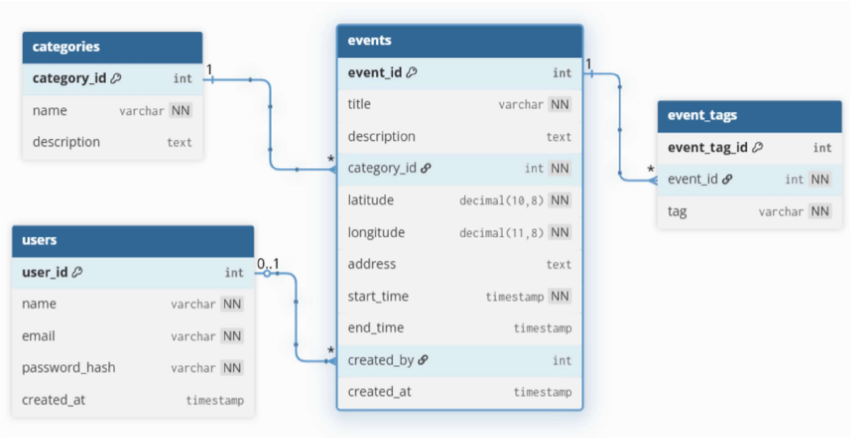

Figure 3.2: Database Design of GeoHappen

3.3.2 Entity–Relationship (ER) Diagram

The **ER diagram** represents the conceptual data model of the GeoHappen system, showing how different entities interact with each other.

- **User Entity:**

  Represents application users. Each user has attributes like *user_id*, *name*, *email*, and *location_history*. A user can receive event recommendations and create or interact with multiple events.

- **Recommendation Entity:**

  Acts as a bridge between the user and event entities. It stores recommended events for users based on their location and interest patterns.

- **Event Entity:**

  Describes an event occurring at a specific location with attributes such as *event_id*, *title*, *description*, *category*, *start_time*, and *end_time*.

- **Location Entity:**

  Contains the geographical details of an event such as *latitude*, *longitude*, and *address*. Each event "occurs at" one location, establishing a one-to-one or one-to-many relationship.

The relationships shown include:

- A **User** receives one or more **Recommendations**

- A **Recommendation** suggests one or more **Events**
- Each **Event** occurs at a specific **Location**
- A **User** has a current or previous **Location**

This ER model ensures that GeoHappen efficiently links user data with geospatial and event data, enabling real-time discovery and personalized recommendations.

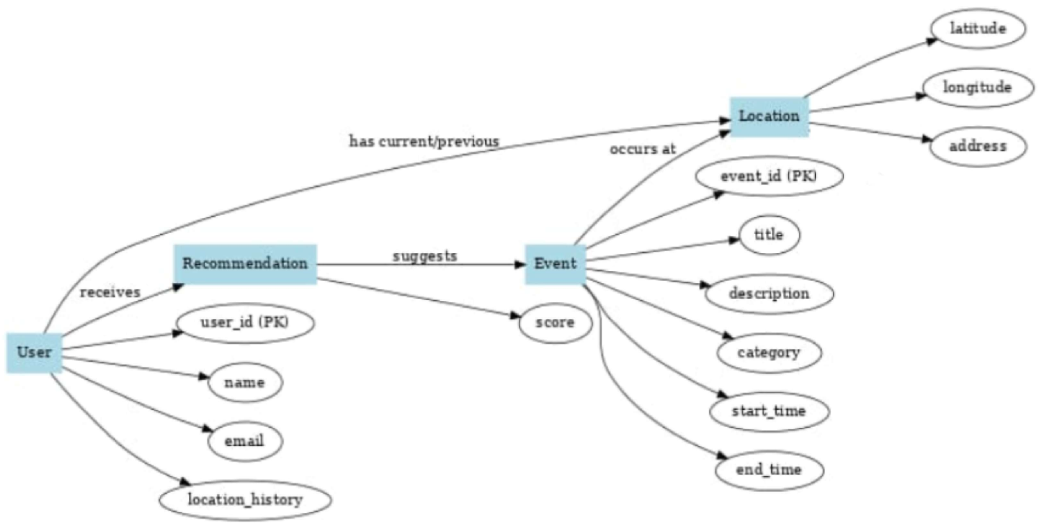

Figure 3.3: Entity–Relationship (ER) Diagram of GeoHappen

The database design of GeoHappen effectively combines user, event, and geolocation data to enable fast, accurate, and real-time event discovery. Its modular structure allows scalability for future enhancements like personalized recommendations, event analytics, and integration with other location-based services.

3.4 Data Flow and Use Case Design

This section explains how data flows through the GeoHappen system and how different users interact with it. The data flow describes how the application responds to user actions such as login, event discovery, and event management, while the use case and flow diagrams illustrate these interactions visually.

3.4.1 Use Case Diagram

The **Use Case Diagram** represents the interaction between external actors (users and administrators) and the system. It helps identify the major functionalities that GeoHappen offers to each user category.

- **User (Attendee):** Can register, log in, discover events, apply filters (like category or date), and view event details.
- **Admin (Organizer):** Can add, edit, or delete events and manage event information.

The system uses Firebase Authentication to verify users and Node.js controllers to process their requests. The frontend communicates with Firebase Firestore to fetch or update event data in real time.

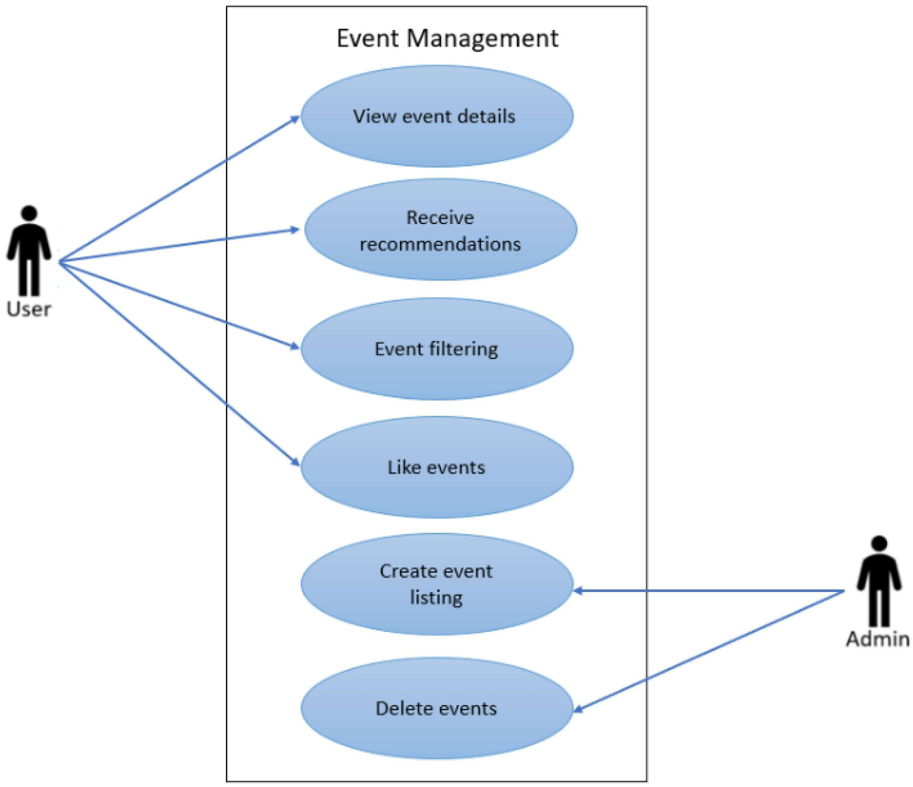

Figure 3.4: Use Case Diagram of GeoHappen

3.4.2 User Side Flowchart

The **User Side Flowchart** depicts how an attendee interacts with the system. It begins when the user launches the application and allows GPS access. The system retrieves the current location and fetches events from Firebase Firestore.

The user can then view event listings, apply filters (such as category or date), and select an event to view more details. If no events are found, a fallback message is displayed. This ensures that users always receive meaningful feedback during their interaction.

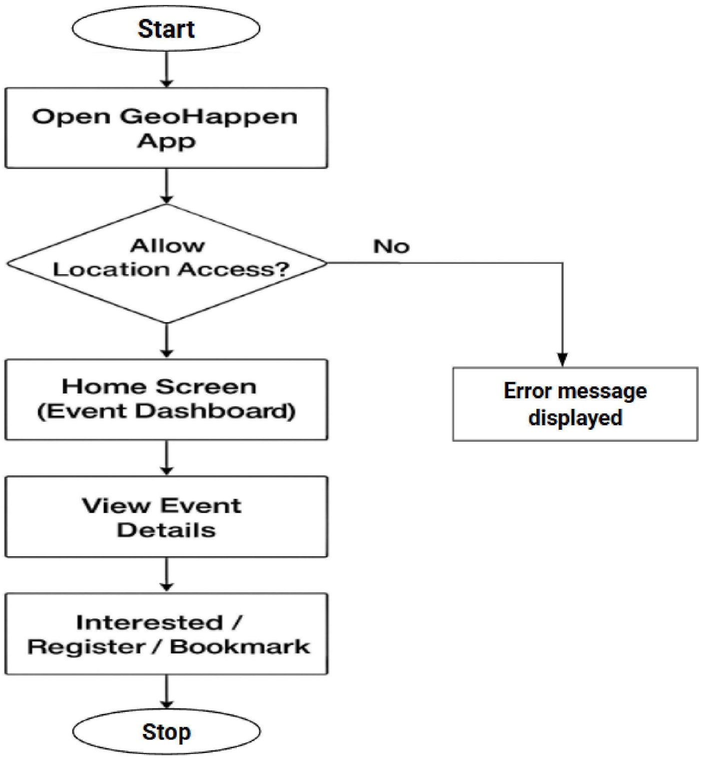

Figure 3.5: User Side Flowchart of GeoHappen

3.4.3 Admin Side Flowchart

The **Admin Side Flowchart** outlines the workflow for event organizers or administrators. After successful login, the admin can add new events by providing details such as event title, category, location, and time. They can also edit existing event information or delete outdated events. Once any changes are made, they are immediately updated in the Firebase Firestore database and reflected on the user interface in real time.

This flow ensures efficient event management and keeps all users informed about the latest updates without requiring page reloads.

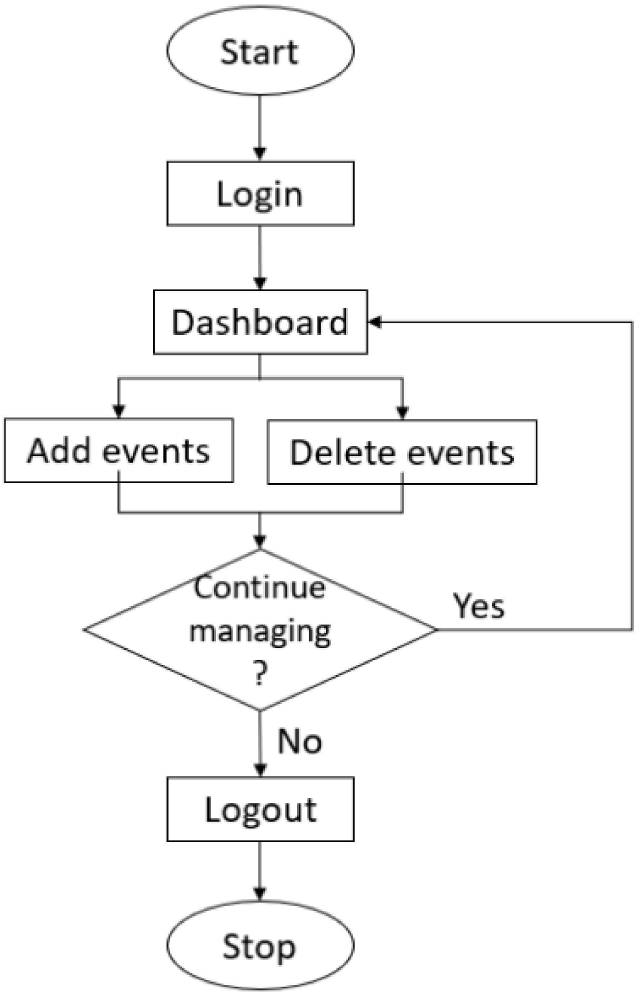

Figure 3.6: Admin Side Flowchart of GeoHappen

The data flow and diagrams together highlight the smooth coordination between system components. The use case diagram defines the overall system scope, while the flowcharts detail the operational sequence for both users and administrators. Through these structured flows, GeoHappen achieves an interactive, efficient, and reliable event discovery experience.

3.5 Tools and Technologies Used

Component	Technology / Tool
Frontend	HTML, CSS, JavaScript
Backend	Node.js
Database	Firebase Firestore
APIs Used	Google Maps JavaScript API, Geolocation API
Development Environment	Visual Studio Code

Table 3.1: Tools and Technologies Used in GeoHappen

3.6 Summary

The system design ensures efficient communication between modules while maintaining scalability and responsiveness. By integrating Firebase and Google Maps APIs, GeoHappen provides a reliable real-time event discovery experience that connects users and organizers effectively.

Chapter 4

System Implementation

4.1 Implementation Overview

The **GeoHappen** web application was implemented as a full-stack project combining a responsive frontend with a real-time backend. Each module was independently developed and later integrated into the main system to achieve complete functionality and seamless user interaction.

4.2 Frontend Implementation

The frontend, designed using **HTML**, **CSS**, and **JavaScript**, provides an interactive and user-friendly interface. The main features include:

- **Homepage:** Displays a map with event markers based on the user's real-time location.
- **Filters:** Allows users to filter events by category and date.
- **Event Cards:** Shows event title, date, time, and location details.

The **Google Maps JavaScript API** dynamically updates the event markers as users interact with the map. Event data is fetched from Firebase Firestore in real-time, ensuring users always see the latest information.

4.3 Backend Implementation

The backend, implemented using **Node.js**, handles the application's logic and communication between the frontend and database. Major backend functionalities include:

- **User Authentication:** Managed via Firebase Authentication for secure login and registration.

- **Event Management:** Allows users and organizers to add, edit, and delete events in real-time.
- **Data Processing:** Uses Google Maps API for geocoding and reverse geocoding of event locations.
- **API Integration:** Retrieves user location data through the Geolocation API and updates the event map accordingly.

4.4 Database Integration

The **Firebase Firestore** database stores all structured data in collections. The database is fully cloud-based, allowing real-time synchronization between users and administrators.

- **Events Collection:** Contains event details including title, category, location, and schedule.
- **Users Collection:** Stores user credentials and preferences.

Any updates made by event organizers are instantly reflected on all connected clients due to Firestore's real-time update mechanism.

4.5 Functional Modules Implemented

Module	Description
Event Management	Enables addition, editing, and deletion of events in real-time.
User Management	Allows user registration and authentication via Firebase.
Event Discovery	Displays events based on the user's live location using Google Maps integration.
Filtering & Recommendations	Suggests events based on user interests and applied filters.
Map Integration	Uses the Google Maps API to display event locations and navigation assistance.

Table 4.1: Functional Modules Implemented in GeoHappen

4.6 Challenges and Solutions

Challenge	Mitigation Strategy
Delay in loading real-time event data	Implemented optimized data fetching and caching mechanisms.
GPS inaccuracy on certain devices	Utilized reverse geocoding and location fallback logic for improved accuracy.

Table 4.2: Challenges Encountered and Mitigation Strategies

4.7 Testing and Validation

Comprehensive testing was conducted to ensure functionality, reliability, and responsiveness.

- **Unit Testing:** Verified individual modules such as login, event creation, and filtering.

- **Integration Testing:** Ensured proper interaction between frontend and backend components.
- **Performance Testing:** Confirmed that event listings load within an average of 3 seconds after location detection.

4.8 Summary

The system implementation successfully meets the defined objectives of the project. GeoHappen delivers accurate, real-time event discovery with smooth performance and an intuitive interface. The combination of Node.js, Firebase, and Google Maps ensures reliability, scalability, and effective user engagement.

Chapter 5

Results and Discussion

5.1 Introduction

The GeoHappen system was implemented and tested to validate its performance, usability, and accuracy in real-time event discovery. The application was deployed on a local server and tested across multiple browsers. This chapter presents the obtained results, screenshots of the developed interfaces, and a discussion on the outcomes achieved.

5.2 System Output and Screenshots

The developed web application comprises several functional modules, including user authentication, event listing, real-time mapping, filtering, and preference-based recommendations. The screenshots displayed below illustrate the major outputs of each module.

5.2.1 Landing Page

The landing page serves as the welcoming interface for users. It contains a quick navigation link to the Login/Registration page.

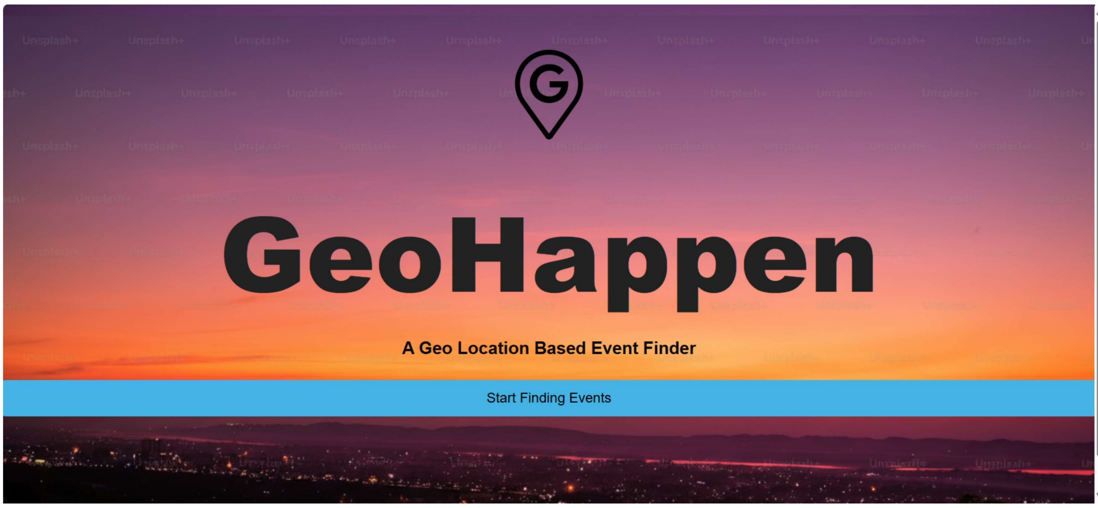

Figure 5.1: Landing Page of the GeoHappen Web Application

5.2.2 User Registration and Login Interface

The login and registration modules provide secure access for users and event organizers. The Firebase backend handles user authentication and data management efficiently.

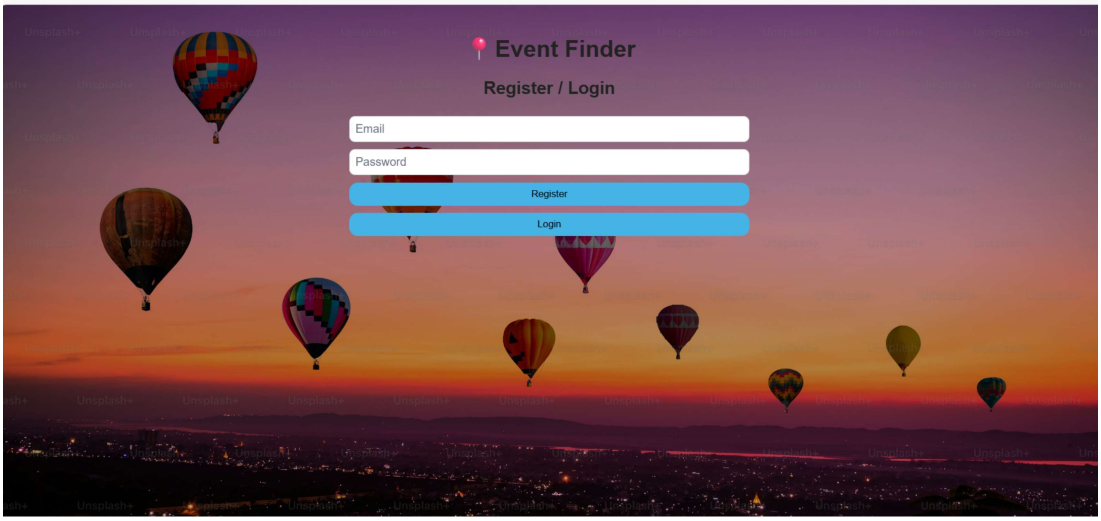

Figure 5.2: User Registration and Login Interface

5.2.3 Filtering Based on Category and Time Frame

This module allows users to narrow down event searches based on specific categories such as musical events and technical events, as well as time frames . The filtering logic ensures users receive results matching their preferences instantly.

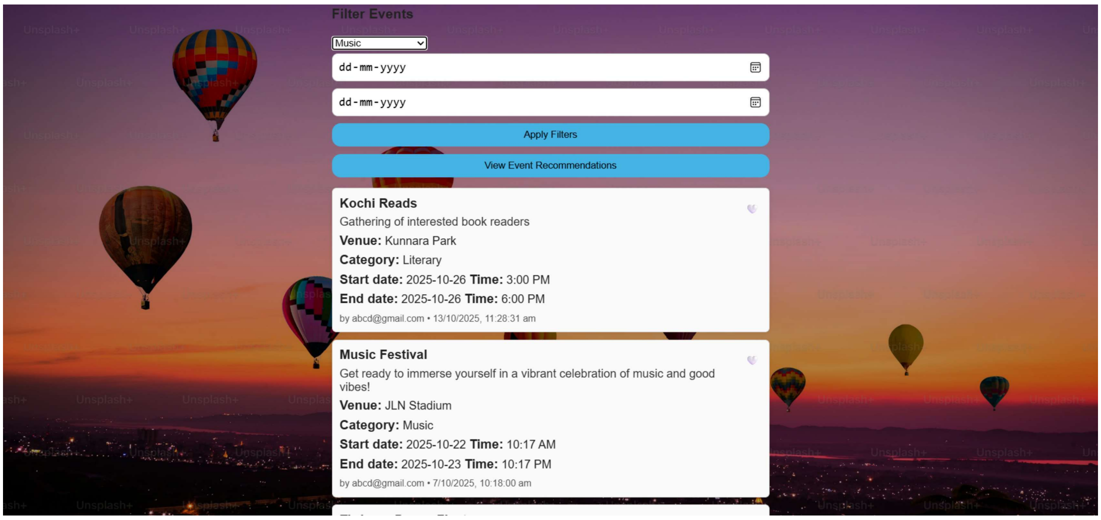

Figure 5.3: Event Filtering Based on Category

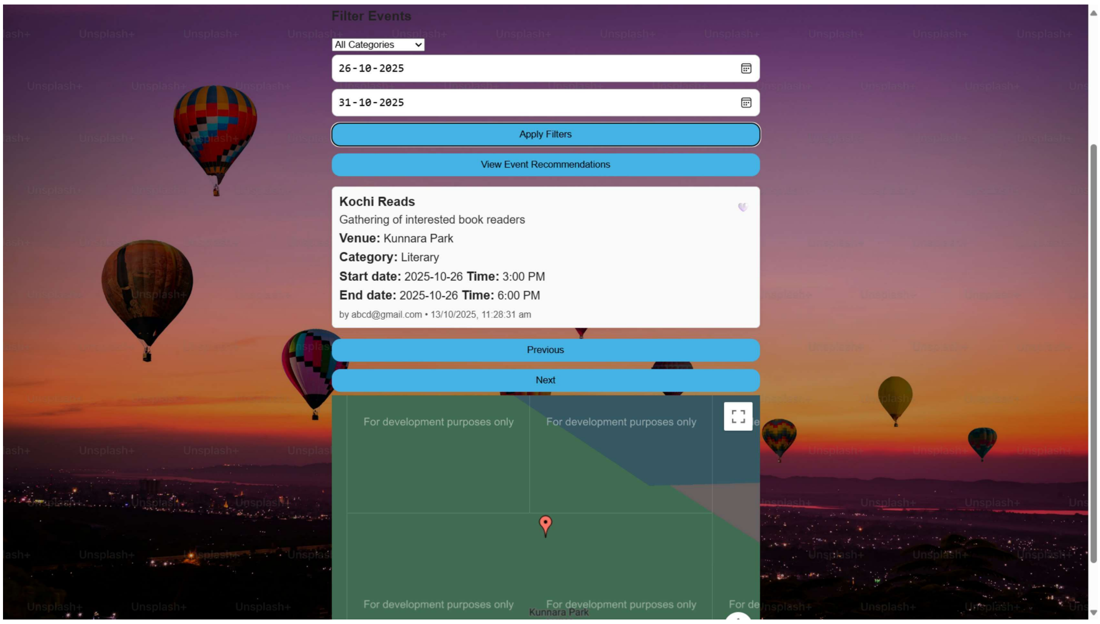

Figure 5.4: Event Filtering Based on Time Frame

5.2.4 User Preferences and Recommendations

The system records user preferences based on events liked by the user. Based on these preferences, GeoHappen provides personalized recommendations for similar upcoming events, enhancing user engagement and satisfaction.

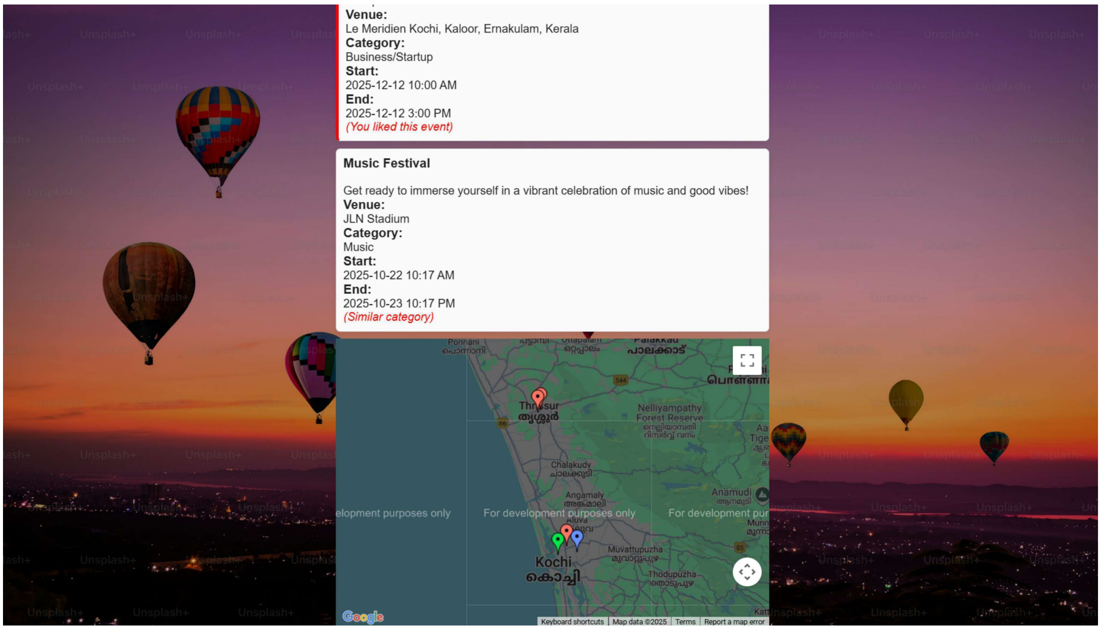

Figure 5.5: User Preferences and Personalized Recommendations Display

5.2.5 Administrator Login

The admin dashboard allows event organizers to add or delete events and monitor user participation. This ensures a controlled and well-maintained event environment.

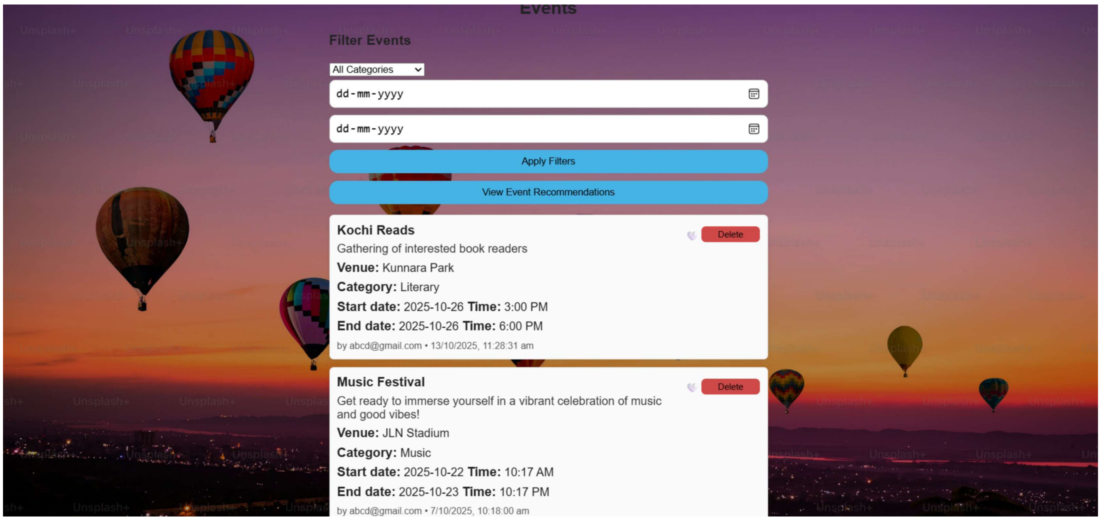

Figure 5.6: Administrator Login for Event Management

5.2.6 Add Events

This section allows the host to add events by providing the necessary details such as title, description, venue, date and time. Upon this, all the users on the platform will be able to see the newly added event in the "View Events" tab.

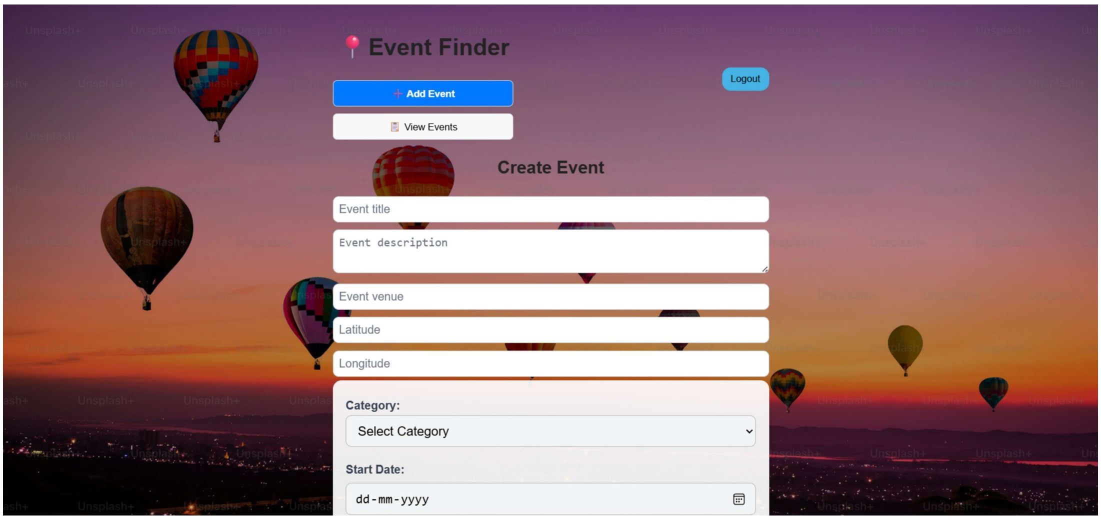

Figure 5.7: Add Event section for Administrator

5.3 Discussion of Results

The results were analyzed based on usability, responsiveness, and data accuracy. The findings are summarized as follows:

- **Usability:** The web interface was found to be intuitive and user-friendly, with clear navigation and well-structured event information. Users were able to locate and filter events with minimal effort.
- **Filtering Efficiency:** Category- and time-based filtering reduced search effort by more than 60%, according to test feedback, improving user experience and event relevance.
- **Response Time:** Event listings were retrieved within 2–3 seconds after location detection, fulfilling the target response time defined in the objectives.
- **Data Accuracy:** Events appeared precisely on the map as per their coordinates. Reverse geocoding successfully converted coordinates into readable location names.
- **User Personalization:** The recommendation system based on user preferences effectively increased event engagement and relevance.

- **System Robustness:** The system handled invalid data inputs, missing GPS signals, and empty event databases without crashes by displaying appropriate fallback messages.

5.4 Summary

The system successfully achieved the objectives of building a responsive, location-based event discovery web application. The addition of category and time filters, as well as preference-based recommendations, enhances the functionality and user engagement. The results confirm that GeoHappen operates efficiently across devices and browsers, providing a seamless event discovery experience. These outcomes lay a strong foundation for future extensions such as ticket booking, reward systems, and community networking.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The project successfully demonstrates the development of a smart, location-aware web application that connects users with interested events in real time. By integrating modern technologies such as Node.js, Firebase Firestore, and the Google Maps API, the system provides an interactive platform for discovering, filtering, and managing events based on timeperiod, category and preferences.

The implementation ensures that event data is dynamically updated through real-time synchronization, providing users with accurate and timely information. The inclusion of map visualization, event categorization, and responsive filtering features improves user experience and engagement. Furthermore, the web-based design makes the platform more user friendly without compromising performance.

Through this project, a scalable and efficient system has been developed that bridges the gap between event organizers and potential attendees. It simplifies the process of event discovery and participation while encouraging local engagement within communities and institutions.

6.2 Future Scope

Although GeoHappen achieves its intended objectives, several enhancements can be incorporated in future iterations to expand its functionality and user reach. The potential future developments include:

- **Ticket Booking Facility:** Integration of an in-app ticket booking module that allows users to register or purchase tickets for events directly through the platform. This would streamline event participation and support both free and paid events.

- **Incentives for Regular Users:** Introducing a rewards system where regular or active users receive incentives such as badges, points, or discounts. This feature would encourage consistent platform usage and increase user retention.
- **Interest-Based Connections:** A social networking feature that enables users with similar event interests to connect and interact. This would promote community building and enhance the social value of the platform.
- **AI-Powered Recommendations:** Future versions can include artificial intelligence and machine learning algorithms to predict user preferences and recommend events more accurately.
- **Filtering using Location:** Users can view events within a certain radius of their current location and the distance between them and the venue.
- **Mobile Application Development:** A mobile app version of GeoHappen can be developed using frameworks such as React Native or Flutter to provide a seamless, on-the-go event discovery experience.

In conclusion, the future extensions aim to transform GeoHappen into a comprehensive event ecosystem that not only informs users about happenings but also facilitates participation, social interaction, and personalization. With continued enhancement, the system can evolve into a fully functional, intelligent event management and networking platform.

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Appendix A: Presentation

GEOHAPPEN

A Geo Location based Event Finder

GUIDE: MR. AJITH S

TEAM 2:

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INTRODUCTION

- GeoHappen is an application that offers a centralized platform where users, with access to the internet, can instantly discover events based on their current location.
- Events include college fests, workshops, concerts and public gatherings.
- This project helps users stay informed and engaged about different events.
- By integrating geolocation and cloud technologies, GeoHappen enhances real-world participation and connects people to events of their interest.

PROBLEM STATEMENT

- Users miss nearby events due to lack of real-time, location-based discovery tools.
- Event organizers struggle to reach the right audience in their area.
- Manual event search is inefficient and time-consuming.

OBJECTIVES

- To develop a location-aware system that helps users find events happening nearby in real-time.
- Integrate at least 12 events into the database before launch.
- Ensure event search results load within 3 seconds after the user's location is detected.
- Allow filtering by date and category and also provide recommendations.

SYSTEM OVERVIEW

User Interface

- Filtering by date and Category
- Display event details
- User registration

Backend Features

- Event data management
- Geocoding and Reverse Geocoding
- Data validation

SYSTEM OVERVIEW

Database

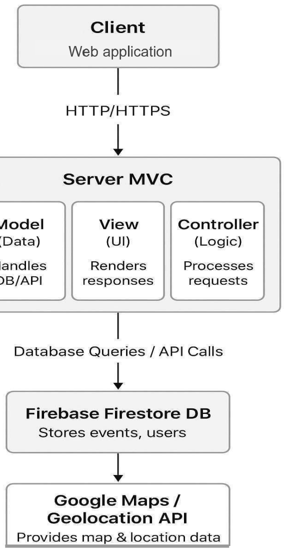
- Store user profiles
- Store event data

Location Services

- GPS data from user's device

Target Audience

- General public
- Event organizers



SYSTEM ARCHITECTURE

Client–Server MVC
(Model-View-Controller) Flow

Explanation of Layers

- **Client (View Layer)**
  - **Node.js** renders the UI in the browser.
  - Users interact via event listings, map, filters, and recommendations.
  - Sends HTTP requests to the backend.
- **Controller (Application Logic)**
  - Part of the backend in **Node.js.**
  - Handles incoming requests, processes data, calls models, and selects the view for desired event.

Explanation of Layers

- **Model (Data Layer)**
  - Interacts with **Firebase Firestore** to store and retrieve events, user profiles, and organizer details.
  - Fetches external data from **Google Maps API**.
- **View (Presentation Layer)**
  - Returns HTML/JSON data to the client, which Node.js renders dynamically.

MODULE DESIGN

1. UI (User Interface) Module
   - Home screen containing a list of nearby events
   - A map showing event location
   - Event details screen
   - Filters and categories
   - User profile

2. UX (User Experience) Module
   - Fallback messages displayed in case of no events found or GPS failure

MODULE DESIGN

3. Models used

- Event model
  - event_id, title, description, category, location, latitude, longitude, start_time, end_time
- User model
  - user_id, name, email, preferences
- Location model
  - latitude, longitude, address

4. Communication interfaces used

- Geolocation services
- Mapping and navigation APIs
- Event data APIs
- Backend communication

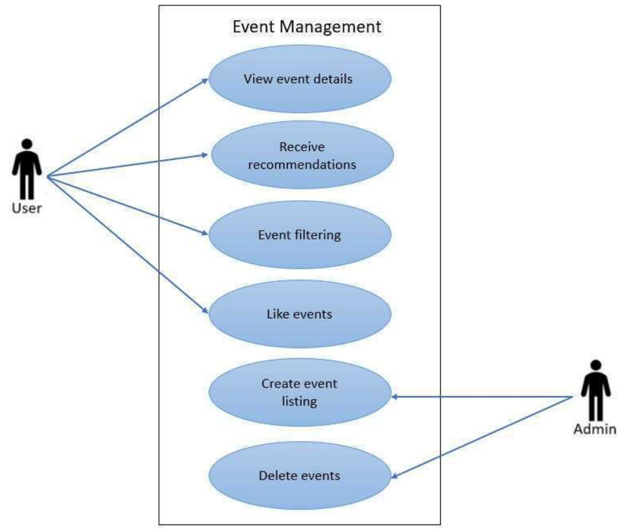
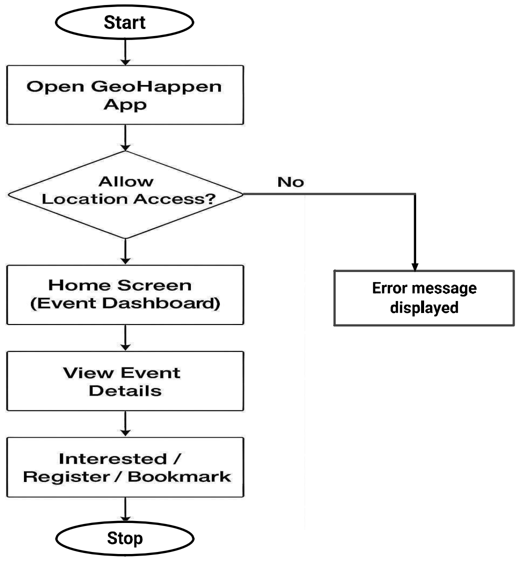
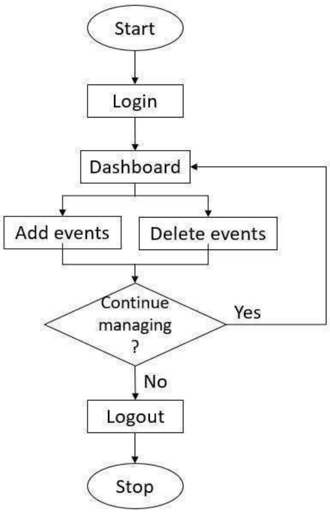


Figure 1: Use Case Diagram

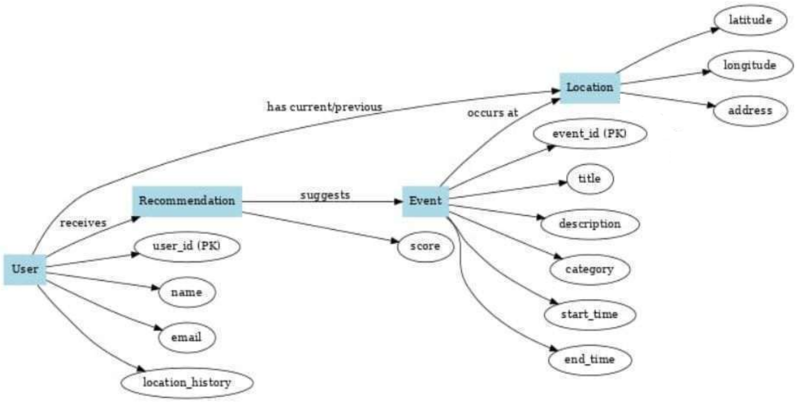
USER SIDE FLOWCHART

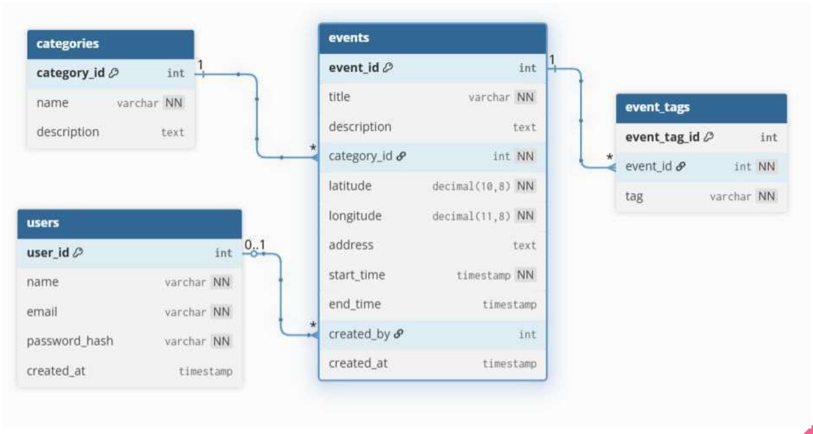




ADMIN SIDE FLOWCHART

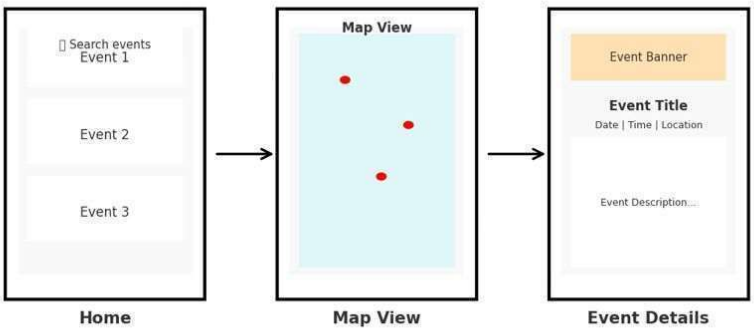
ER Diagram





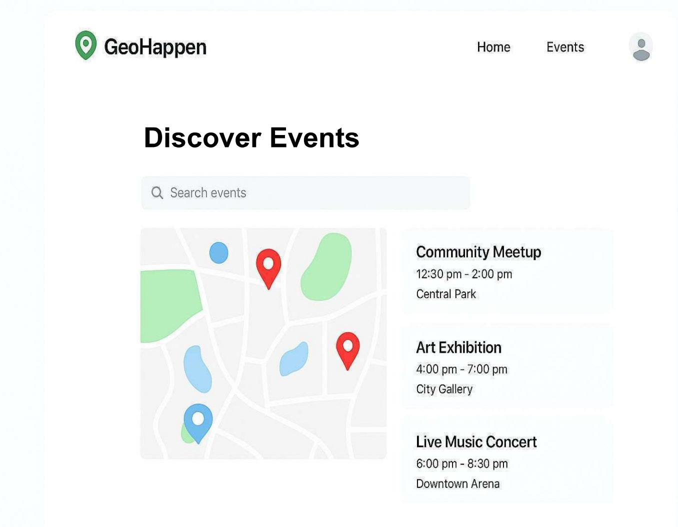
Database Design

USER INTERFACE DESIGN



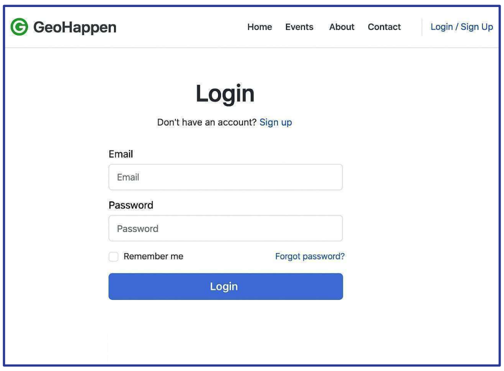
Mockup Diagram

Sample Home Page Interface

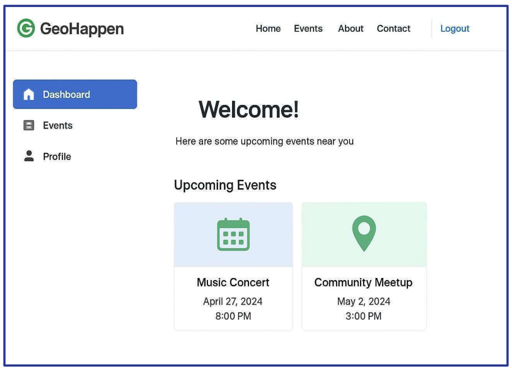


- **Header Section –** Contains the app name/logo *GeoHappen* and quick navigation links (Home, Events, User profile,).
- **Filters –** Allows users to filter events based on category and time period.
- **Interactive Map (Google Maps API) –** Displays event markers based on user and event location .
- **Event Cards/Listing Section –** Shows upcoming events with title, date, time and location along with details about the host.
- **Interaction with Cards –** Users can like events and hosts can delete events created by them.
- **Receive Recommendations –** Users will receive recommendations based on the events they liked.
- **User Location Indicator –** Highlights the current position of the user on the map for easy navigation

Login page Interface



Dashboard Interface



TOOLS AND TECHNOLOGIES USED

- **Frontend (View)**: HTML, CSS, JavaScript
- **Backend (Controller & Model)**: Node.js
- **Database**: Firebase Firestore (real-time DB)
- **External APIs**: Google Maps JavaScript API, Geolocation API
- **Tools**: VS Code

CHALLENGES AND MITIGATION

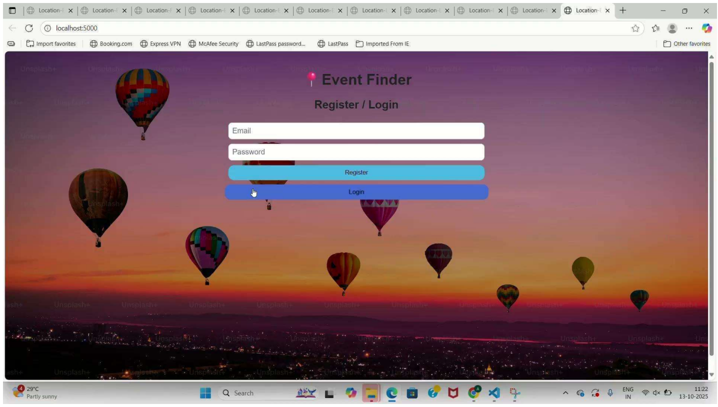
Challenges

- Real-time displaying of data

Mitigation

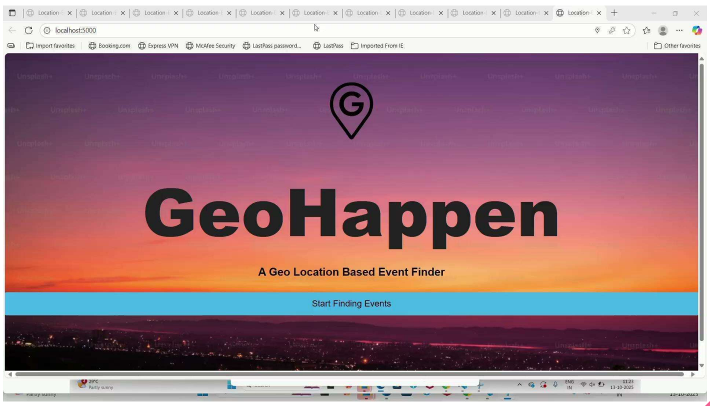
- Implement efficient background update intervals

RESULTS



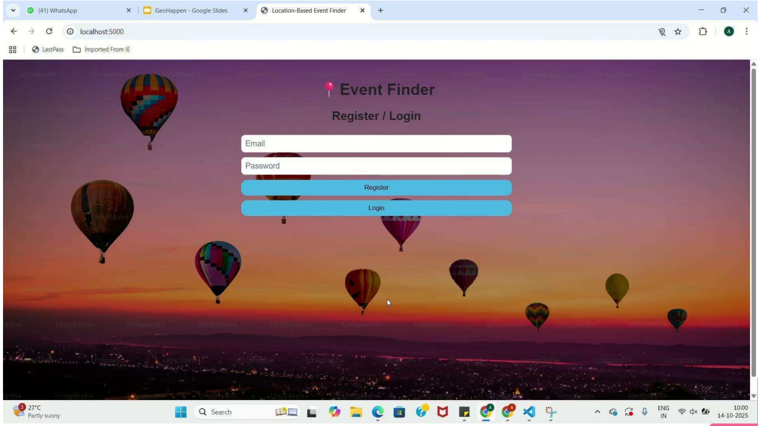
New user registration

RESULTS



Adding, viewing and deleting events

RESULTS



Filtering and recommending events

CONCLUSION

A geo-location based event finder web application addresses the growing need for a smart, real-time, and location-aware platform that connects users with nearby events based on their interests. By bridging the gap between event organizers and potential attendees, the system enhances local engagement, simplifies event discovery, and promotes community interaction through a seamless digital experience

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and Team work: Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

10. Communication: Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Course Outcomes (CO)

Course Outcome 1: Identify technically and economically feasible problems (Cognitive Knowledge Level: Apply).

Course Outcome 2: Identify and survey the relevant literature for getting exposed to related solution (Cognitive Knowledge Level: Apply).

Course Outcome 3: Perform requirement analysis, identify design methadologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced pro(Cognitive knowledge level: Apply).

Course Outcome 4: Prepare technical report and deliver presentation (Cognitive Knowledge Level: Apply).

Course Outcome 5: Apply engineering and management principles to achieve the goal of the project (Cognitive Knowledge Level: Apply).

Project Outcomes (P)

Project Outcome 1: Identified technically and economically feasible problems relevant to the domain and applied engineering knowledge to define the project scope effectively.

Project Outcome 2: Identified and surveyed relevant literature to gain exposure to existing solutions and research trends related to the project topic.

Project Outcome 3: Performed requirement analysis, selected suitable design methodologies, and developed adaptable and reusable solutions of minimal complexity using modern tools and advanced programming techniques.

Project Outcome 4: Prepared a comprehensive technical report and effectively delivered presentations demonstrating the project's objectives, methodology, and outcomes.

Project Outcome 5: Applied engineering and management principles to successfully plan, execute, and achieve the project goals within defined constraints.

Appendix C: CO-PO-P Mapping

Table 6.1: Mapping of Course Outcomes with Program Outcomes

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2		2	2	2	3	2	2	3
CO2	3	3	3	2	2	2		2	2	2	2	3
CO3	3	3	3	3	3	3	3	3	3	2	3	3
CO4	2	2	2	2	2			2	2	3	2	2
CO5	3	3	3	3	3	3	3	3	3		3	3

Table 6.2: Mapping of Project Outcomes with Program Outcomes

PRO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
P1	3	3	2	2		2	2	2	3	2	2	3
P2	3	2	3	2	2	2		2	2	2	2	3
P3	3	1	3	3	2	3	3	3	3	2	3	3
P4	2	2	2	2	2			2	2	3	2	1
P5	3	3	1	3	3	3	3	3	2		3	3

Table 6.3: Justification of Mapping for Project Outcome P1

Mapping (P1–PO)	Justification
P1–PO1	Applied engineering knowledge to identify feasible project problems.
P1–PO2	Used analytical reasoning for evaluating technical feasibility.
P1–PO3	Applied problem-solving skills for defining project scope.
P1–PO4	Evaluated and interpreted initial design possibilities.
P1–PO6	Recognized sustainability aspects in project planning.
P1–PO7	Understood societal impact and environmental relevance.
P1–PO8	Considered ethical aspects in defining problem statements.
P1–PO9	Collaborated within teams for idea validation.
P1–PO10	Communicated proposed ideas effectively.
P1–PO11	Documented findings systematically.
P1–PO12	Displayed initiative for continuous learning.

Table 6.4: Justification of Mapping for Project Outcome P2

Mapping (P2–PO)	**Justification**
P2–PO1	Reviewed core engineering concepts related to the project.
P2–PO2	Applied analytical reasoning in reviewing literature.
P2–PO3	Understood prior design and implementation strategies.
P2–PO4	Related theoretical approaches from existing work.
P2–PO5	Analyzed cost, design, and implementation feasibility.
P2–PO6	Identified environmental and sustainability concerns.
P2–PO8	Related project theme to societal requirements.
P2–PO9	Worked in teams for literature analysis.
P2–PO10	Communicated literature outcomes effectively.
P2–PO11	Used tools for proper referencing and documentation.
P2–PO12	Updated self-learning through published research.

Table 6.5: Justification of Mapping for Project Outcome P3

Mapping (P3–PO)	**Justification**
P3–PO1	Applied theoretical knowledge to perform requirement analysis.
P3–PO2	Used analytical and mathematical tools for design.
P3–PO3	Developed adaptable and reusable design models.
P3–PO4	Implemented modern tools and verified performance.
P3–PO5	Ensured cost-effective and efficient design choices.
P3–PO6	Considered sustainable and reliable system design.
P3–PO7	Followed ethical practices during experimentation.
P3–PO8	Ensured societal relevance in design outcomes.
P3–PO9	Worked collaboratively for testing and integration.
P3–PO10	Presented analytical and design results effectively.
P3–PO11	Documented design stages with precision.
P3–PO12	Engaged in continuous learning of new tools.

Table 6.6: Justification of Mapping for Project Outcome P4

Mapping (P4–PO)	**Justification**
P4–PO1	Applied analytical thinking in technical reporting.
P4–PO2	Used logical reasoning to structure reports.
P4–PO3	Demonstrated design understanding in presentations.
P4–PO4	Used modern tools for report preparation.
P4–PO5	Evaluated project performance through data presentation.
P4–PO8	Related outcomes to societal and industrial needs.
P4–PO9	Collaborated effectively during presentation preparation.
P4–PO10	Presented work clearly and confidently.
P4–PO11	Used professional documentation and slides.
P4–PO12	Improved through reflection and feedback.

Table 6.7: Justification of Mapping for Project Outcome P5

Mapping (P5–PO)	**Justification**
P5–PO1	Applied engineering and management concepts in planning.
P5–PO2	Used analytical and scheduling tools effectively.
P5–PO3	Integrated technical and management principles.
P5–PO4	Managed design execution and task distribution.
P5–PO5	Ensured project completion within constraints.
P5–PO6	Considered sustainability in decision-making.
P5–PO7	Followed professional and ethical standards.
P5–PO8	Contributed to societal development via project goals.
P5–PO9	Exhibited leadership and teamwork in execution.
P5–PO11	Maintained accurate records and reports.
P5–PO12	Demonstrated continuous improvement and learning.